ANANYA SAHA
MTECH SCHOLAR

ananya2002saha@gmail.com Website
•orcid.org/0009-0001-2209-4971 •GitHub: github.com/yourusername
•Google Scholar: Link •LinkedIn: Link

RESEARCH INTERESTS

Medical Image Analysis, Attention Mechanisms and Model Explainability, Few-Shot Learning, Multi-Modal Fusion and Meta-Learning, Clinical Diagnostics AI. Broad interest in translating advanced machine learning techniques into practical diagnostic tools for resource-limited healthcare settings.

EDUCATION

MTech in CSE (Artificial Intelligence and Data Science Spcialization)
•Jadavpur University •2024-2026 •Grade(First Year): 9.12 CGPA

BTech in CSE
•Techno Main SaltLake •2020-2024 •Grade: 9.38 CGPA

Higher Secondary School (XII)
•Indira Gandhi Memorial High School •2019-2020 •Grade: 92%

Secondary School (X)
•Julien Day School, Kalyani •2017-2018 •Grade: 93.75%

ACADEMIC QUALIFICATIONS

NET ,Dec 2025, CSE
Qualified: PHD, Assistant Professor

GATE ,2024, CSE
Qualified: 591 score

YGPA ,2023-2024, (4th Year)
Position: 1, YGPA:9.45

National Cyber Olympiad ,2017
City Rank: 2, Zonal Rank(WB): 78
International Rank : 549

RESEARCH EXPERIENCE

ONGOING

March 2026 **GILF-Net: Gated Iterative Learnable Lifting Fusion for Dual-Branch Attention-Based Medical Image Classification**
Supervisor: Prof. Ram Sarkar, JU
Developed a dual-encoder deep learning architecture with a novel gated iterative lifting fusion mechanism for medical image classification. The framework combines MobileNetV2-SE and DenseNet201-CBAM encoders, projects features into a shared 512-dimensional latent space via 1×1 convolutions, and performs bidirectional iterative refinement using learnable residual updates and spatial gating. The fusion module dynamically exchanges complementary information between branches across T iterations, progressively enhancing discriminative representations. Evaluated on LAG (glaucoma), HAM10000 (skin lesions), and PBC (blood cells) datasets, achieving superior accuracy over single-branch baselines and conventional fusion methods.

Feb 2026 **GI-GEM: Geometry-Influenced Explainability for Deep Vision Models**
Supervisor: Prof. Ram Sarkar, JU
Proposed GI-GEM (Geometry-Influenced Gradient Explanation Maps), a model-agnostic explainability framework that analyzes geometric sensitivity of deep vision models using controlled affine and elastic transformations. The method fuses geometric influence maps with gradient-based activations to generate structure-aware explanations without retraining. Evaluated on ImageNet, 3DIRCADb, and JUVDsi v1, the approach produces sharper localization and improved faithfulness compared to Grad-CAM and LIME across multiple vision tasks.

Feb 2026 **Contrastive Training and Multi-Prompt Distillation based Few-Shot Learning for Skin Lesion Classification**
Supervisor: Prof. Ram Sarkar, JU
Proposed a three-stage few-shot learning framework for skin lesion classification under limited labeled data. The pipeline integrates contrastive image–text pretraining, prompt-consistency distillation with cross alignment and knowledge distillation with frozen teacher, and prototypical few-shot learning using a CLIP ViT-B/32 backbone to learn domain-aware visual–semantic representations. Evaluated on HAM10000, ISIC 2018, and BreakHis, the method achieves strong performance across multiple few-shot settings (1–10 shots), demonstrating improved robustness for medical image classification in data-scarce scenarios.

ACCEPTED/PUBLISHED

April 2026 **Few-Shot Learning based Multimodal Skin Lesion Classification**
Authors: Saptarshi Pani*, Ananya Saha*, Ram Sarkar [*equal contributions]
ACCCEPTED in PHAROS-AIF-MIH Workshop, held in conjunction with *CVPR 2026*
Developed a dual-stream few-shot learning framework integrating visual and textual modalities for skin lesion classification under limited labeled data. The vision stream combines frozen CLIP ViT-B/16 and DenseNet-121 encoders with a Gaussian fuzzy adaptive fusion using learnable confidence weights based on activation magnitudes. Designed a novel alignment mechanism between prompt-tuned CLIP text encoder (learnable context tokens) and a trainable metadata encoder, projecting both into shared CLIP embedding space for logit-level fusion. Evaluated on PAD-UFES-20, HAM10000, and ISIC datasets in 1–10 shot settings, achieving competitive accuracy with reduced supervision.

March 2026 **CoMFDNet: A Correlation-guided Meta-Fusion based Dual Network for Glaucoma Classification**
Authors: Ananya Saha, Gouranga Maity, Ram Sarkar
PUBLISHED in NHIB (Springer)
Proposed CoMFDNet, a dual-network framework for automatic glaucoma classification from retinal fundus images. The model combines ResNet50 and MobileNetV2 feature extractors with a cosine similarity–based correlation fusion module, enhanced by channel and spatial attention mechanisms. A meta-learner adaptively balances both branches to improve robustness. Evaluated on ACRIMA and LAG, the approach achieves up to 99.63% accuracy, demonstrating strong performance for automated clinical glaucoma screening.

June 2025 **NeuralCodOpt: Codon optimization for the development of DNA vaccines**
Authors: Tapan Chowdhury, Aishwarya Saha, Ananya Saha, Arnab Chakraborty, Nibir Das
PUBLISHED in Computational Biology and Chemistry (Elsevier)
Developed NeuralCodOpt, a machine learning–based algorithm for organism-specific codon optimization to improve gene translation efficiency. The framework combines K-Means clustering to estimate target Codon Adaptation Index ranges and reinforcement learning to automate sequence optimization. Evaluated across 130 species, the method achieves 86.7% accuracy and demonstrates potential to enhance protein expression in DNA expression vectors, supporting applications in gene therapy and DNA vaccine development.

May 2023 **Commercial aspect of Telemedicine in Indian scenario**
Authors: Prof. Jafor Ali Akhan, Ananya Saha, Shubhankar Ghosh, Dr. Himadri Nath Saha
PUBLISHED in IJESI
Analyzed the evolution and impact of telemedicine in healthcare delivery, particularly in low- and middle-income countries such as India. The study highlights the expansion of SATCOM-based telemedicine networks initiated by Indian Space Research Organisation and the rapid adoption of telehealth services during the COVID-19 pandemic by the Ministry of Health and Family Welfare. Findings emphasize telemedicine's potential to improve healthcare accessibility and reduce system burden in resource-constrained settings.

March 2023 **Heart Disease Treatment in Telemedicine using Internet of Things**
Authors: Prof. Jafor Ali Akhan, Ananya Saha, Shubhankar Ghosh, Dr. Himadri Nath Saha
PUBLISHED in IJESI
Presented at the 5th Regional Science and Technology Congress, Region-4 (Kolkata, Howrah, South 24 Parganas), 2022–23.
[Abstract published in the Conference Book of Abstracts]
Proposed a low-cost Internet of Things–based telemedicine system for real-time heart disease monitoring. The framework uses a fingertip-based heart rate sensor to detect and transmit patient heartbeat signals wirelessly to a computer or Android application for visualization and analysis. The system enables continuous remote monitoring, alerts, and feedback to help detect potential cardiac risks and improve telemedicine-based healthcare management.

SKILLS

- **Programming & Development:** Python, C, Web Development (HTML, CSS, JavaScript)
- **Emerging Technologies:** Machine Learning, Deep Learning, Internet of Things (IoT)
- **Research & Academic Skills:** Literature Review, Scientific Writing, Content Writing
- **Professional Skills:** Teamwork, Leadership, Laboratory Assistance, Student Mentoring
- **Languages:** Bengali (Native/Fluent), English (Professional Proficiency), Hindi (Limited Proficiency)

TEACHING/MENTORING EXPERIENCE

Present ↑ March 2026

Lab Assistant, of lab UG/CSE/ES/S/123 (Computer Programming Lab)
Jadavpur University
Assisted Prof. Ram Sarkar in conducting Computer Programming laboratory sessions for two groups of 1st-year B.Tech CSE students (~60 students total, 30 per group). Delivered supplementary lectures on core C programming concepts, administered weekly programming assignments, and provided live debugging and problem-solving support during lab sessions. Evaluated student code implementations and maintained attendance records.

Present ↑ Oct 2025

Student Mentor, CMATER Lab
Jadavpur University
Mentored undergraduate research intern under Prof. Ram Sarkar's supervision. Guided training in transfer learning implementations, multi-modal fusion techniques, and deep learning frameworks (PyTorch/TensorFlow). Provided instruction on research methodology including literature review, experimental design, result analysis, and academic writing. Supervised hands-on projects involving medical image classification and model evaluation, fostering development of technical and scholarly skills for graduate-level research.

Dec 2025 ↑ Aug 2025

Lab Assistant, of lab CSE/PC/B/S/311 (Computer Graphics Lab)
Jadavpur University
Assisted Prof. Ram Sarkar in conducting Computer Graphics laboratory sessions for three groups of 3rd-year B.Tech CSE students (~90 students total, 30 per group). Delivered supplementary lectures on core graphics concepts, designed and administered weekly programming assignments, and provided live debugging and problem-solving support during lab sessions. Evaluated student code implementations, maintained attendance records, and invigilated internal laboratory examinations. Supervised and mentored student mini-projects on graphics algorithms and visualization techniques.

REFERENCES

Prof. Ram Sarkar
Associate Professor of CSE
Jadavpur University
ram.sarkar@jadavpuruniversity.in

Dr. Tapan Chowdhury
Associate Professor of CSE
Techno Main Salt Lake
tapan2005cse@gmail.com

Dr. Himadri Nath Saha
HOD of CSE
SNEC, Calcutta University
Contactathimadri@gmail.com